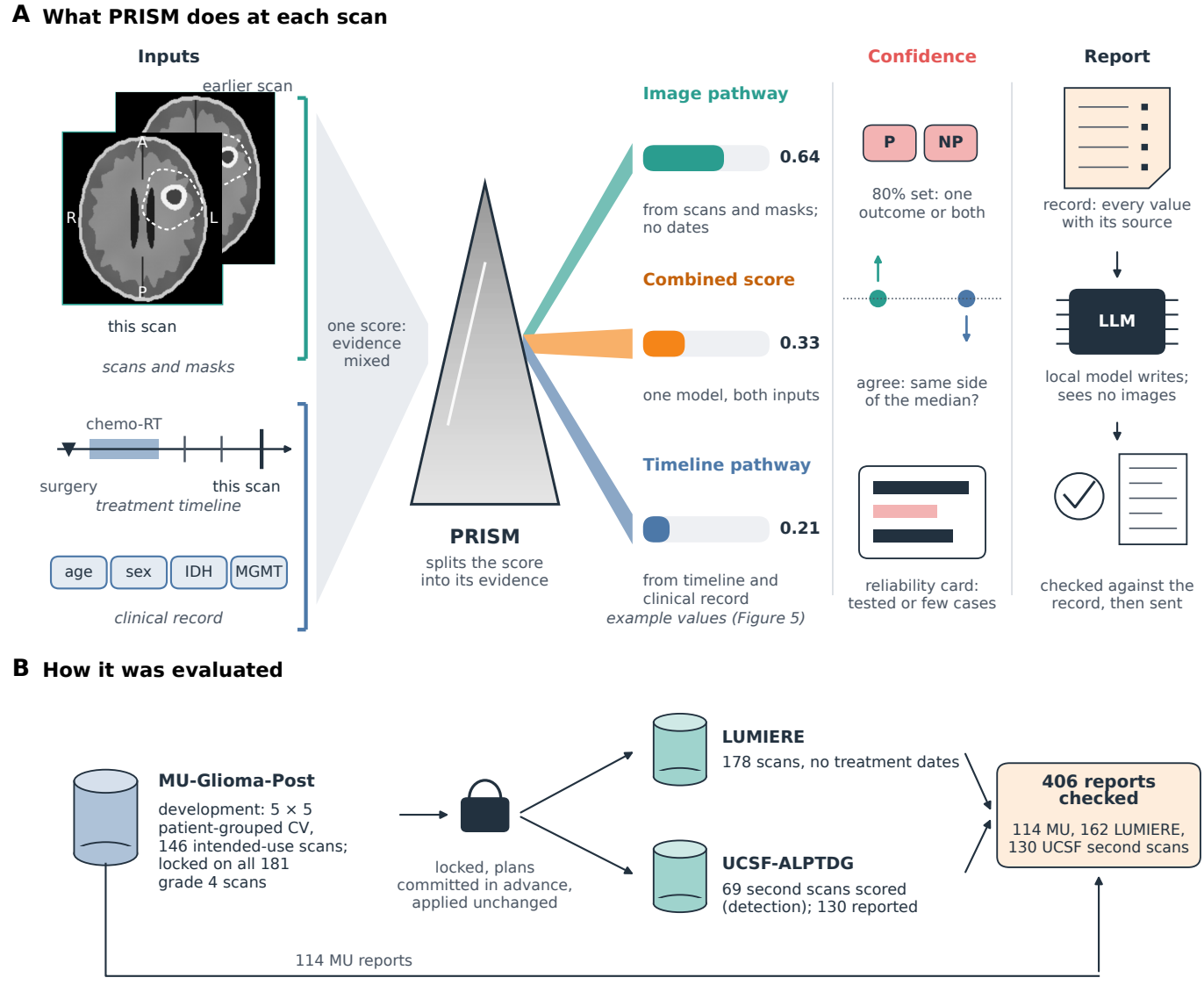

A What PRISM does at each scan
Inputs
earlier scan
A
R
L
P
this scan
scans and masks
chemo-RT
surgery
this scan
treatment timeline
age
sex
IDH
MGMT
clinical record
one score: evidence mixed
PRISM
splits the score into its evidence
Image pathway
0.64
from scans and masks; no dates
Combined score
0.33
one model, both inputs
Timeline pathway
0.21
from timeline and clinical record
example values (Figure 5)
Confidence
P
NP
80% set: one outcome or both
agree: same side of the median?
reliability card: tested or few cases
Report
record: every value with its source
LLM
local model writes; sees no images
checked against the record, then sent
B How it was evaluated
MU-Glioma-Post
development: 5 × 5 patient-grouped CV, 146 intended-use scans; locked on all 181 grade 4 scans
locked, plans committed in advance, applied unchanged
LUMIERE
178 scans, no treatment dates
UCSF-ALPTDG
69 second scans scored (detection); 130 reported
406 reports checked
114 MU, 162 LUMIERE, 130 UCSF second scans
114 MU reports